Michael Bailey

AI Engineer — LLM Applications, Python & TypeScript

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PROFESSIONAL SUMMARY

Software engineer with 3 years at Sky, now leading AI adoption in the Global Reliability team. Built solo an AI-powered self-serve dashboard that replaces a BA-mediated change-request workflow consuming ~95 engineer-days/year: users describe a chart in natural language and drop returned widgets into their own dashboards, backed by a custom Model Context Protocol (MCP) server and agent orchestrator over OpenAI GPT. Previously owned Sky's internal monitoring dashboards — Next.js, GraphQL, OpenSearch — used across multiple departments. Software Developer Level 4 apprenticeship, Distinction.

KEY SKILLS

AI & LLM Applications: LLM application development, OpenAI (GPT) models, agentic workflow design, Model Context Protocol (MCP), custom agent orchestration, tool use / function calling, prompt engineering, streaming responses (SSE), embeddings, vector search (OpenSearch)

Languages: Python, TypeScript (confident), JavaScript · Java, Golang, C# (familiar)

Frameworks & Platforms: Next.js, React, Node.js, FastAPI, Express, GraphQL, Recharts

Data & Search: OpenSearch (search, analytics, vector), MySQL, SQL

Cloud: Google Cloud Platform (GCP)

CI & Tooling: Jenkins (CI/CD), Docker, pytest, Jest; deployed on Sky's internal FluidCloud platform

Ways of Working: Technical leadership of AI initiatives · running internal AI onboarding sessions · production incident response (Global Reliability) · stakeholder demos across Sky teams and to Comcast executives

EXPERIENCE

Software Engineer — Sky UK

Sep 2022 – Present · London · Global Reliability

AI initiatives led:

- Designed and built solo an AI-powered self-serve monitoring dashboard: users ask questions like *"show me 4G usage split by tech in a stacked bar chart over the last eight months"* and a custom agent returns chart widgets they can resize, edit in-place, and assemble into shareable, per-user dashboards — collapsing a BA-mediated change-request cycle of 1-2 days per chart into sub-minute self-serve.
- Architected a custom MCP server (TypeScript) and agent orchestrator (Python, uvicorn) to work around tool-calling limits in Sky's internal AI gateway — wiring OpenSearch data sources, a Next.js / React / TypeScript frontend, and OpenAI GPT models into a single tool-use loop, Dockerised and deployed via Jenkins on Sky's FluidCloud platform.
- Authored the compliance sign-off business case — covering security posture (prompt-injection boundaries, handle-based result retrieval to prevent raw-row leakage, tenant isolation), efficiency modelling (~35% projected gain against the ~95-day baseline), cost projection (~£100/month run-cost), and phased rollout.
- Ran 3 stakeholder demos during pre-rollout hardening: Sky's data engineering team, another Sky observability team, and a group of Comcast executives (Sky's parent company).
- Leading a department-wide shift to agentic workflows at my director's request — running onboarding sessions on using AI effectively and safely across engineering and non-engineering work.

Core engineering:

- Shipped an earlier reliability-insights tool for Sky's on-call rota — embedding-based retrieval over OpenSearch surfaces similar past incidents during triage, shortening time-to-context on repeat-class failures.

- Owned a suite of full-stack internal monitoring dashboards used across multiple Sky departments for 3 years (Next.js / React / TypeScript frontend, GraphQL middleware, OpenSearch data) — the direct foundation for the AI-powered self-serve dashboard above, re-using the same data contract.

SIDE PROJECTS

Ask F1 — Conversational F1 dashboard + race-replay viewer

Live: f1-dashboard-bay.vercel.app · Source: github.com/MBailey114/f1-dashboard

Public version of the Sky self-serve pattern, rebuilt on the Vercel AI SDK to learn the primitives end-to-end. Two surfaces, one Next.js process, no agent framework.

- **Dashboard** — chat agent with 6 typed Zod tools (`query_f1`, `add_widget`, `remove_widget`, `replace_widget`, `relayout_dashboard`, `compose_dashboard`) that mutate a 12-column grid in real time. Single prompt → 4–6 widget themed dashboard via one streamed tool call. Custom agent step timeline (auto-collapses to a pipeline pill, expands to per-tool input/output inspectors with mini-cards / column-strip / source-citation renderers).
- **Race replay viewer** (`/tracks`) — pre-baked 2025 lap telemetry (OpenF1 location + car_data, 6 curated laps) replayed at 60Hz with Catmull-Rom interpolated car motion, clickable mini-sector arc segments, multi-driver speed/throttle/brake chart, driver-portrait hover popovers, and a live AI race-commentary HUD (gpt-5.4-nano, topic-rotation + entity-blocklist logic to prevent repetition across a 30-line lap).
- **Cross-cutting** — Cmd+K command palette, global keyboard shortcuts, shareable URLs (LZ-compressed widget state, plain query params for tracks), F1 broadcast-style design system, driver-mention chips throughout chat.

Three OpenAI model tiers: gpt-5.5 (agentic dashboard chat), gpt-5.4 (scoped widget/track chat with Tavily web search), gpt-5.4-nano (suggestions, race commentary). Edge runtime on every API route with Next's data cache for cross-invocation persistence; deployed on Vercel Hobby.

Stack: Next.js 16, React 19, Vercel AI SDK v6, TypeScript strict, Recharts 3, Tailwind 4, Zustand 5 (with persist + selector subscriptions), Zod 4, d3-geo, lz-string, cmdk. Pure TypeScript pivot logic — model never sees the records array, only field names plus three sample rows.

EDUCATION

Software Developer (Level 4) Apprenticeship — Multiverse / Sky UK

Sep 2022 – Apr 2024 · Distinction

Completed the Software Developer Level 4 apprenticeship standard (Institute for Apprenticeships and Technical Education) while working full-time in Sky's engineering organisation. Training provider: Multiverse. End-point assessment awarded Distinction on 24 April 2024. Work-based project: Sky's internal monitoring dashboards (Next.js / React, GraphQL, OpenSearch).